MILITARY SPECIFICATION

FERRIC OXIDE, TECHNICAL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two classes of ferric oxide.

1.2 Classification. The ferric oxide shall be of the following classes as specified in the contract or order (see 6.2):

Class 1 - Fine
Class 2 - Extra Fine

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

OD 18903	Iron or Iron as Ferric Oxide, Determination of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of the invitation for bids or the request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Wasington, DC 20462. Orders should specify "49 CFR 100-199 (latest revision)").

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material shall be ferric oxide in powder form.

3.2 Chemical and physical properties. The chemical and physical properties shall be in accordance with TABLE I.

TABLE I. Chemical and physical properties.

Property	Class 1		Class 2	
	Minimum	Maximum	Minimum	Maximum
Moisture, wt%	-	0.5	-	0.5
Calcination loss, wt% (dry basis)	-	2.0	-	2.0
Iron, as Fe_2O_3, wt% (calcined basis)	98.0	-	98.0	-
Acidity, as H_2SO_4, wt%	-	0.02	-	0.02
Weight Mean Diameter Particle Size, Microns	4.5	6.0	2.0	3.0

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3.3 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)

b. Altitude: Normal ground

c. Vibration: None

d. Humidity: Room ambient to 95 percent relative, maximum

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all material manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time. Several batches, manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.3.2 Sampling. Sampling for quality conformance inspection shall be in accordance with inspection level I of MIL-STD-105. The sample unit shall be one unit package or container of material. Each sample shall consist of sufficient material to perform the quality conformance tests as specified in 4.4.

4.4 Quality conformance inspection. Each sample obtained in accordance with 4.3.2 shall be subjected to the tests of 4.5. The acceptable quality level (AQL) shall be 2.5% defective. When specified in the contract or order (see 6.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests specified for each lot of material.

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4.5 Tests. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Moisture content. The moisture content shall be determined in accordance with the following:

a. Transfer a sample of approximately 2 grams (g) to a tared 30 millilitre (ml) silica crucible. Cover the crucible with a tight-fitting tared cover and weigh to the nearest 0.1 mg.

b. Remove the cover and place the crucible in an oven at 105 to 110°C for 1 hour, minimum.

c. Replace the cover, remove the crucible from the oven, cool in a desiccator and weigh to the nearest 0.1 mg. Retain covered sample for use in the calcination loss test of 4.5.2.

d. Calculate the weight percent (wt%) moisture as follows:

$$\text{Moisture, wt\%} = \frac{(A - B)}{(A - C)} \times 100$$

where: A = Weight of crucible, cover and sample, g
B = Weight of crucible, cover and dried sample, g
C = Weight of crucible and cover, g

e. Report the results of 2 determinations and their average.

4.5.2 Calcination loss. The calcination loss shall be determined on a dry basis in accordance with the following:

a. Remove the cover from the crucible with sample from 4.5.1c and place the crucible in a muffle furnace at 800 to 900°C for 30 minutes, minimum.

b. Remove the crucible from the furnace and allow to cool for 2 to 3 minutes. Replace the cover, place the crucible in a desiccator and cool to room temperature. Weigh the covered crucible to the nearest 0.1 mg. Retain covered sample for use in the iron content test of 4.5.3.

c. Calculate the calcination loss on a dry basis as follows:

$$\text{Calcination loss, wt\%} = \left[\frac{B - D}{B - C}\right] \times 100$$

where: B = Weight of crucible, cover and dried sample (from 4.5.1e), g
C = Weight of crucible, cover and ignited sample, g
D = Weight of crucible and cover (from 4.5.1e), g

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d. Report the results of 2 determinations and their average.

4.5.3 Iron content. The iron content as weight percent Fe_2O_3 shall be determined on a calcined basis in accordance with 4.5.3.1 or 4.5.3.2. In the event of controversy the results obtained by the method of 4.5.3.1 shall be determinative.

4.5.3.1 Preferred method for iron content. The preferred method for determination of iron as weight percent Fe_2O_3 shall be in accordance with the following:

a. Prepare or obtain the following reagents:

1. Hydrochloric acid, concentrated
2. Stannous chloride solution - prepare fresh by dissolving 12.5 g, weighed to the nearest 0.1 mg, of stannous chloride in 25 ml hot concentrated hydrochloric acid and diluting to 250 ml with distilled water.
3. Mercuric chloride, saturated solution - dissolve 100 g of mercuric chloride in 1 litre of distilled water by heating.
4. Sulfuric-phosphoric acid mixture - made by adding 150 ml concentrated sulfuric acid to 500 ml distilled water, adding 150 ml of 85 percent phosphoric acid and diluting to 1 litre with distilled water.
5. Barium diphenylamine sulfonate indicator, 0.2 percent.
6. Standard potassium dichromate solution, 0.1 normal(N).

b. Weight 0.28 to 0.32 g of ignited sample from 4.5.2b to the nearest 0.1 mg into a 400 ml beaker.

c. Add 20 ml of concentrated hydrochloric acid and cover the beaker. Heat on a hot plate until the sample is completely dissolved. Remove the cover and rinse the cover and the sides of the beaker with 15 to 20 ml of distilled water.

d. With the beaker still on the hot plate, slowly add stannous chloride solution dropwise until the last drop makes the solution colorless or free from any tinge of yellow, then add an additional 1 or 2 drops. Remove the beaker from the heat and rapidly cool to room temperature. Quickly add 10 ml of mercuric chloride solution, swirl a few seconds until uniform in appearance, then set aside for 2 or 3 minutes to allow a slight white precipitate to form. If precipitate does not form or if a heavy grayish precipitate forms, discard and perform another determination using more or less stannous chloride, as indicated.

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e. If precipitate forms correctly, add 200 ml of water, 20 ml of sulfuric-phosphoric acid mixture, and 6 to 8 drops of 0.2% barium diphenylamine sulfonate indicator. Titrate with standard potassium dichromate solution, taking the sudden change from gray-green to blue as the endpoint. Run a blank determination with the same reagents.

f. Calculate the wt% Fe_2O_3 on a calcined basis as follows:

$$\text{Iron as } Fe_2O_3, \text{ wt\%} = \left[0.07985 \frac{(V-v)N}{W}\right] X\ 100$$

where: V = Volume of potassium dichromate solution used to titrate sample, ml
v = Volume of potassium dichromate solution used to titrate blank, ml
N = Normality of potassium dichromate solution
W = Weight of ignited sample used, g
0.07985 = Milliequivalent weight of Fe_2O_3

g. Report the results of 2 determinations and their average.

4.5.3.2 Alternate method for iron content. The alternate method for determination of iron as wt% Fe_2O_3 shall be in accordance with OD 18903 except that a calcined sample shall be used.

4.5.4 Acidity. The acidity as wt% H_2SO_4 shall be determined in accordance with the following:

a. Weigh 10 ± 0.5 g of sample to the nearest 0.1 mg. Add 100 ml of distilled water and boil for several minutes.

b. Filter and rinse sample with hot distilled water, adding rinsings to the filtrate.

c. Titrate the filtrate with 0.1N potassium hydroxide using methyl orange indicator.

d. Run a blank determination.

e. Calculate the acidity as wt% H_2SO_4 as follows:

$$\text{Acidity as } H_2SO_4, \text{ wt\%} = \left[\frac{0.049\ (V-v)N}{W}\right] X\ 100$$

where: V = Volume of potassium hydroxide solution used to titrate sample, ml
v = Volume of potassium hydroxide solution used to titrate blank, ml

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N = Normality of potassium hydroxide solution
W = Weight of sample, g
0.049 = Milliequivalent weight of sulfuric acid

f. Report the results of a minimum of 2 determinations and their average.

4.5.5 Weight mean diameter. Weight mean diameter shall be determined in accordance with the following:

a. Reagent: Nitrogen, prepurified.

b. Apparatus:

1. Sharples Micromerograph, type XC or equivalent.
2. Templates, 6 inches/minute and 6 inches/hour.
3. Stopwatch.

c. Cleaning procedure: Clean the micromerograph in accordance with the manufacturers instructions.

d. Instrument settings: Set the instrument for the ferric oxide in accordance with Table II.

TABLE II. Instrument Settings.

DEAGGLOMERATOR OPENING (MICRONS)	NITROGEN PRESSURE (PSI)	SENSITIVITY POSITION	APPROXIMATE SAMPLE SIZE (mg)
250	100	1	45

e. Procedure:

1. Remove the sample holder by releasing the clamping handle and pulling straight out.
2. With a micro-spatula, scoop a small portion of the material into the sample holder. (See Table II for approximate sample size).
3. Replace sample holder into the feed chamber and clamp it into position.
4. Set deagglomerator opening, feed pressure and sensitivity (See Table II).
5. Normally the power to the micromerograph is left on. This keeps the machine in a ready condition and avoids warm-up time.

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6. With the recorder switch on, push down on the chart speed lever. The down position drives the chart at 6 inches/minute and the up position drives the chart at 6 inches/hour.
7. With the chart moving at 6 inches/minute move the recorder pen to the zero position on the chart, using the zero adjuster at the bottom of the recorder.
8. Press the left hand charging button for 5 seconds filling the pressure chamber.
9. Press the right hand firing button and hold down for 5 seconds. This discharges the powder into the sedimentation column.
10. After the powder has been discharged into the column, let 225 seconds elapse and then change the chart speed to 6 inches/hour by pushing the chart speed lever upward.
11. Allow the run to continue until the pen deflection becomes constant. The micromerograph run is then completed.
12. Turn off the recorder and open the sedimentation column by turning the seal operating handle counterclockwise. Place the balance pan protector disc over the balance pan.
13. If a total deflection of 60 is not obtained, the run is discarded and the sample is rerun.
14. Remove the chart from the recorder and identify the chart as to material, lot number, instrument settings, date and any other pertinent information.

f. Data reduction and calculations:

1. Lay the template marked 6 inches/minute over the chart, aligning it with the change of speed point and align the lines parallel to the sides of the template with the 0 to 100 scale on the outer edge of the recorder chart for the full length of the template. Read the chart deflection at the intersection of the template arc lines and the chart curve, recording the data on the calculation sheet in column 3 of Table III.
2. Lay the template marked 6 inches/hour over the chart, aligning it with the change of speed point and align the lines parallel to the sides of the template with the 0 to 100 scale on the outer edge of the recorder chart for the full length of the template. Read the chart deflection at the intersection of the template arc lines and the chart curve, recording the data on calculation sheet in column 3.
3. Normalize all figures in column 3 to 100 percent by dividing each by the last figure in column 3, and multiply by 100 to convert to percentage, and record in column 4.

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TABLE III. Calculation sheet, weight mean diameter of Ferric Oxide.

1	2	3	4	5
$d\sqrt{p}$	Diameter, microns	Chart deflection	Cumulative % greater than	Cumulative % less than
269	164			
180	110			
150	91.5			
120	73.2			
90	54.9			
75	45.7			
66	40.2			
60	36.6			
54	32.9			
48	29.3			
42	25.6			
36	22.0			
30	18.3			
27	16.5			
24	14.6			
21	12.8			
20	12.2			
19	11.6			
18	11.0			
17	10.4			
16	9.76			
15.1	9.21			
14	8.54			
12	7.32			
10	6.10			
8	4.88			
6	3.66			
5	3.05			
4	2.44			
3	1.83			
2.5	1.52			
2	1.22			

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4. Column 2 has been calculated by dividing each figure in column 1 by the square root of the density of aluminum.
5. Subtract each entry in column 4 from 100 and list the result in column 5.
6. Plot cumulative weight percent less than diameter (column 5) as the ordinate (arithmetic scale) against diameter, in microns, (column 2) as the abcissa (log scale) on 3 cycle semi-log graph pater. Identify the plot as described in 4.5.5.e.14 .
7. The weight mean diameter, in microns, is reported as the intercept of the 50 percent line and the curve plotted in 6. above.

4.6 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order (see 6.2), packaging and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract or order (see 6.2), packaging and packing of ferric oxide shall be in accordance with standard commercial practice applicable to the type of material. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract or order (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179. Marking shall include, but not be limited to, the following information:

a. Title, number and date of this specification

b. Manufacturer's name and location

c. Material trade name

d. Net weight

e. Lot number, batch number(s), and date of manufacture

f. Storage conditions

g. Contract or purchase order number

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6. NOTES

6.1 Intended use. Ferric oxide in accordance with this specification is intended for use as an ingredient of solid propellant. Class 1 ferric oxide is specifically intended for use in the CCU-22/A impulse cartridge assembly. Class 2 ferric oxide is specifically intended for use in the CKU-5/A rocket catapult assembly.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification

b. Quantity required

c. Place of delivery

d. Inspection conditions when other than as specified (see 4.2)

e. Lot size if other than as specified (see 4.3.1)

f. Packaging requirements if other than as specified (see 5.1)

g. Size of container required (see 5.1.1)

h. Any special markings required (see 5.2)

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Quality conformance inspection data	-

*DIDs (Data Item Description/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Suggested source of supply. Products that have met the requirements of this specification in past procurement actions are RY-2196 and R-5098, for Class 1 and Class 2 respectively, manufactured by Charles Pfizer and Company, 324 Harbor Way, so San Francisco, CA 94080.

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This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Custodian:
Navy - OS

Preparing Activity:
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Project Number:
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